Technical Service Bulletin

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High Horsepower Aluminum Piston Debond Failures	

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Core Issue

The purpose of this document is to:

- Aid in correctly identifying aluminum piston debond failures for high horsepower engines.
- Aid in determining the correct root cause of failure for aluminum piston debond failures on high horsepower engines.

A piston debond failure is defined as a condition where the cast iron ring carrier in the piston comes loose from the base aluminum piston.

Confirmation

Product Affected :

- K19 (All Versions)
- K38 (All Versions)
- K50 (All Versions)

See Service Bulletin, Operation of Diesel Engines in Cold Climates, Bulletin 3379009 (/qs3/pubsys2/xml/en/bulletin/3379009.html), when diagnosing the root cause of an aluminum piston debond failure.

Unacceptable engine performance and even engine damage can occur if the required items in Service Bulletin 3379009 are **not** implemented.

In addition, Cummins Inc. recommends consulting a local Cummins® authorized repair location for regional specific information regarding the operation of diesel engines in cold climates.

Resolution

Metallurgical analysis of aluminum pistons has revealed that a piston debond failure can occur as a result of the following:

- Higher than normal cylinder temperature combustion.
- Mechanical overload.
- Improper cold climate operation.

Some cold climate operation issues that can cause aluminum piston debond failures are:

- Un-metered ether injection. Un-metered ether is defined as the use of ether from a spray can rather than a device that controls the amount of fluid injected into the air intake manifold.
- Failed or improperly functioning pre-heaters.
- Excessive idle times. Excessive idle time is defined as greater than 15 minutes.
- Non-temperature controlled fan drive, such as a viscous or thermatic fan drive.
- Inadequate sealing of the engine compartment from cold air blasts.
- Intake air not drawn from engine compartment in cold weather operation.
- Lubricating oil heaters not being utilized.

See Service Bulletin, Operation of Diesel Engines in Cold Climates, Bulletin 3379009 (/qs3/pubsys2/xml/en/bulletin/3379009.html), for more detailed information on the above issues and additional items that may be needed

Piston Failure Analysis

The condition of an aluminum piston that has experienced a debond failure can help determine the potential root

cause of the failure.

Cracks on the combustion bowl could indicate excessive heat and/or over-loading from:

- Un-metered ether use
- Excessive idle times
- High air inlet restriction
- High exhaust back pressure
- Injection timing too advanced for engine operating condition
- Low intake air temperatures at low engine speeds or idle

Piston burning or pitting from severe detonation could be caused by:

- Excessive idle times
- Low intake air temperatures when operating at low engine speeds or idle.
    - Both conditions above can cause fuel puddles to form in the crown of the piston and/or above the top piston ring. The fuel puddles ignite under a higher operating load and cause piston material erosion or removal and premature piston material softening.

Piston crown meltdown could be caused by:

- Over-fueling
    - The over-fueling may not be a problem with the fuel system. A restricted air intake system or exhaust system can reduce the air charge in the cylinder. The fuel system will try to “make up” for the lack of power because of poor air and exhaust circulation by adding fuel.
- A blocked, restricted or improperly aimed piston cooling nozzle.

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Document History

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